ADSP 31014 Statistical Models for Data Science
Course Project Part 1

(20 points)

Objective

Find an interesting data set of your choice. Perform data wrangling, cleaning, exploratory data analysis, and visualization. Write a (up to) 3-page report to describe the nature of the data, characteristics of the column variables, any interesting aspect(s) or anomaly, and include selected informative visualizations. Readers of your report should acquire a good sense of your data set after reading it.

Deliverables

- Fit all your write-up onto 3 pages. No more.
- The report should be in .pdf or .docx or similar document file format.
- No code in the report itself. Just text and plots.
- Submit both the report and any related reproducible code (in R or Python) to Canvas.

Some Considerations

- Later in Course Project Part 2, you will define a business problem based on this data set and apply either linear regression, logistic regression, or Poisson regression modeling. Therefore, your data set should have a sizable number of rows (e.g. $n \geq 1000$) and at least one (numerical for linear regression, binary or multi-class categorical for logistic regression, or non-negative integer count for Poisson regression) column as the response variable y.
- It might make your life easier and the business problem more meaningful if the columns that you might use as explanatory variables / predictors are at least somewhat related to the response variable. In this way, you avoid the ugly situation of null result i.e. "nothing has a relationship with anything else".
- A categorical column (even if it has a lot of categorical levels i.e. high cardinality) might still be OK to work with e.g. grouping them.
- If you are unsure about the suitability of your data set, please talk to the instructor.

Copyright © 2026 Danny Ng